

42136 - project part 2

April 9, 2026

- Hand in date: April 22nd at 23:59.
- The project should be made in groups of 2 to 3 persons
- The answer to exercise 1 should at most be 2 pages long.
- The answer to exercise 2 should at most be 2 pages long.
- Hand in all your Julia code.

1 Exercise 1

Task 1.1

Consider the following IP

$$\begin{array}{ll} \max & x + 2y \\ \text{s.t.} & x + y \geq 2 \quad (A1) \\ & x + y \leq 5 \quad (A1) \\ & x - y \geq 0 \quad (A1) \\ & x - y \leq 1 \quad (A1) \\ & y \leq 3 \quad (A0) \\ & x - y \geq -1 \quad (A0) \\ & x \leq 4 \quad (A0) \\ & \frac{1}{3}x + y \geq \frac{7}{3} \quad (A0) \\ & (x, y) \in \mathbb{Z}^2. \end{array}$$

Visualize the feasible region of the IP and its LP relaxation in a 2D coordinate system

Task 1.2

Now we wish to write up the Dantzig-Wolfe relaxation of the problem by “convexifying” the A1 constraints together with the integrality constraints. Touch upon the following in your answer:

- What are the extreme points of X_1 (convex hull of integer solutions to the A1 constraints)?
- Show the Dantzig-Wolfe relaxation of the problem (an LP model).
- Solve the Dantzig-Wolfe relaxation version of the problem.
- Illustrate the solution to the Dantzig-Wolfe relaxation together with the IP solution and LP relaxation of the original problem using the figure from task 1.2

2 Exercise 2

Here we consider a variant of the Generalized assignment problem where some jobs are incompatible in the sense that they cannot be served together on the same machine. We use the following notation.

- I : set of machines
- J : set of jobs
- c_i : capacity of machine i
- w_{ij} : capacity used when assigning job j to machine i
- p_{ij} : profit of assigning job j to machine i
- $K = \{(a_1, b_1), (a_2, b_2), \dots, (a_\kappa, b_\kappa)\}$: set of incompatible jobs. For each pair $(a_k, b_k) \in K$ we must ensure that jobs a_k and b_k are not planned on the same machine
- $x_{ij} \in \{0, 1\}$: Variable. 1 if we assign job j to machine i , 0 otherwise

The mathematical model is:

$$\max \sum_{i \in I} \sum_{j \in J} p_{ij} x_{ij}$$

subject to

$$\sum_{i \in I} x_{ij} = 1 \quad \forall j \in J \quad (1)$$

$$\sum_{j \in J} w_{ij} x_{ij} \leq c_i \quad \forall i \in I \quad (2)$$

$$x_{ia} + x_{ib} \leq 1 \quad \forall (a, b) \in K, i \in I \quad (3)$$

$$x_{ij} \in \{0, 1\} \quad \forall i \in I, j \in J \quad (4)$$

An instance is provided on DTU learn (exercise2.data.jl).

2.1 Task 2.1

Implement the model and solve the instance from the data set.

2.2 Task 2.2

We consider the Dantzig-Wolfe decomposition we get by keeping constraint (1) in the master problem and “convexifying” constraints (2), (3) and (4).

- Solve the Dantzig-Wolfe relaxation using the Automatic column generation/Dantzig-Wolfe decomposition code from lecture 3.

2.3 Task 2.3

Adapt the model from week 3, exercise 3 to the generalized assignment problem with incompatible jobs. Explain shortly what changes in the model. Adapt your code from week 3, exercise 3 to solve the LP relaxation of the model (which should be equivalent to the Dantzig-Wolfe relaxation from task 2.2). Does the solution match that of task 2.2?